



TAFICITA

Emtricitabine and Tenofovir Alafenamide Tablets
200 mg/25 mg

1. **NAME OF THE MEDICINAL PRODUCT**
TAFICITA (Emtricitabine and Tenofovir Alafenamide Tablets 200 mg/25 mg)
2. **QUALITATIVE AND QUANTITATIVE COMPOSITION**
Each tablet contains 200 mg of Emtricitabine and Tenofovir Alafenamide Fumarate equivalent to 25 mg of tenofovir alafenamide.
3. **PHARMACEUTICAL FORM**
Film-coated tablet.
A blue, film-coated, rectangular, biconvex, beveled edge tablet debossed with M on one side of the tablet and EAI on the other side. The film-coated tablet should not be chewed, crushed, or SPLIT
- CLINICAL PARTICULARS

- 4.1 **Therapeutic indications**
TAFICITA is indicated in combination with other antiretroviral agents for the treatment of adults and adolescents (aged 12 years and older with body weight at least 35 kg) infected with human immunodeficiency virus type 1 (HIV-1) (see sections 4.2 – Posology & Method of Administration and 5.1 Pharmacodynamic Properties).
- 4.2 **Posology and method of administration**
Therapy should be initiated by a physician experienced in the management of HIV infection.
- Posology**
Adults and adolescents aged 12 years and older, weighing at least 35 kg
TAFICITA should be administered as shown in Table 1.
- Table 1: Dose of TAFICITA according to third agent in the HIV treatment regimen**

Dose of TAFICITA	Third agent in HIV treatment regimen (see section 4.5)
Emtricitabine and Tenofovir Alafenamide 200/10 mg once daily	Atazanavir with ritonavir or cobicistat Darunavir with ritonavir or cobicistat ¹ Lopinavir with ritonavir
Emtricitabine and Tenofovir Alafenamide 200/25 mg once daily	Dolutegravir, efavirenz, maraviroc, nevirapine, rilpivirine, raltegravir

1. Emtricitabine and Tenofovir Alafenamide 200/10 mg in combination with darunavir 800 mg and cobicistat 150 mg, administered as a fixed-dose combination tablet, was studied in treatment naive subjects, see section 5.1.
If the patient misses a dose of TAFICITA within 18 hours of the time it is usually taken, the patient should take TAFICITA as soon as possible and resume the normal dosing schedule. If a patient misses a dose of TAFICITA by more than 18 hours, the patient should not take the missed dose and simply resume the usual dosing schedule.
If the patient vomits within 1 hour of taking TAFICITA another tablet should be taken.

- Elderly**
No dose adjustment of TAFICITA is required in elderly patients (see sections 5.1 and 5.2).
- Renal impairment**
No dose adjustment of TAFICITA is required in adults or adolescents (aged at least 12 years and of at least 35 kg body weight) with estimated creatinine clearance (CrCl) ≥ 30 mL/min. TAFICITA should be discontinued in patients with estimated CrCl that declines below 30 mL/min during treatment (see section 5.2).

- No dose adjustment of TAFICITA is required in adults with end stage renal disease (estimated CrCl < 15 mL/min) on chronic haemodialysis; however, TAFICITA should generally be avoided but may be used in these patients if the potential benefits are considered to outweigh the potential risks (see sections 4.4 and 5.2). On days of haemodialysis, TAFICITA should be administered after completion of haemodialysis treatment.
TAFICITA should be avoided in patients with estimated CrCl ≥ 15 mL/min and < 30 mL/min, or < 15 mL/min who are not on chronic haemodialysis, as the safety of TAFICITA has not been established in these populations.
No data are available to make dose recommendations in children less than 18 years with end stage renal disease.
- Hepatic impairment**
No dose adjustment of TAFICITA is required in patients with hepatic impairment.
- Paediatric population**
The safety and efficacy of TAFICITA in children younger than 12 years of age, or weighing < 35 kg, have not yet been established. No data are available.

- Method of administration**
TAFICITA should be taken orally, once daily with or without food (see section 5.2). The film-coated tablet should not be chewed, crushed, or split.
- 4.3 **Contraindications**
Hypersensitivity to the active substances or to any of the excipients listed in section 6.1.
- 4.4 **Special warnings and precautions for use**
While effective viral suppression with antiretroviral therapy has been proven to substantially reduce the risk of sexual transmission, a residual risk cannot be excluded. Precautions to prevent transmission should be taken in accordance with national guidelines.
- Patients co-infected with HIV and hepatitis B or C virus**
Patients with chronic hepatitis B or C treated with antiretroviral therapy are at an increased risk for severe and potentially fatal hepatic adverse reactions.
The safety and efficacy of TAFICITA in patients co-infected with HIV-1 and hepatitis C virus (HCV) have not been established.
Tenofovir alafenamide is active against hepatitis B virus (HBV). Discontinuation of TAFICITA therapy in patients co-infected with HIV and HBV may be associated with severe acute exacerbations of hepatitis. Patients co-infected with HIV and HBV who discontinue TAFICITA should be closely monitored with both clinical and laboratory follow-up for at least several months after stopping treatment.

- Liver disease**
The safety and efficacy of TAFICITA in patients with significant underlying liver disorders have not been established (see sections 4.2 and 5.2).
Patients with pre-existing liver dysfunction, including chronic active hepatitis, have an increased frequency of liver function abnormalities during combination antiretroviral therapy (CART) and should be monitored according to standard practice. If there is evidence of worsening liver disease in such patients, interruption or discontinuation of treatment must be considered.
- Weight and metabolic parameters**
An increase in weight and in levels of blood lipids and glucose may occur during antiretroviral therapy. Such changes may in part be linked to disease control and life style. For lipids, there is in some cases evidence for a treatment effect, while for weight gain there is no strong evidence relating this to any particular treatment. For monitoring of blood lipids and glucose reference is made to established HIV treatment guidelines. Lipid disorders should be managed as clinically appropriate.
- Mitochondrial dysfunction following exposure in utero**
Nucleos(t)ide analogues may impact mitochondrial function to a variable degree, which is most pronounced with stavudine, didanosine and zidovudine. There have been reports of mitochondrial dysfunction in HIV negative infants exposed *in utero* and/or postnatally to nucleoside analogues; these have predominantly concerned treatment with regimens containing zidovudine. The main adverse reactions reported are haematological disorders (anaemia, neutropenia) and metabolic disorders (hyperlactatemia, hyperlipaemia). These events have often been transitory. Late onset neurological disorders have been reported rarely (hypertonia, convulsion, abnormal behaviour). Whether such neurological disorders are transient or permanent is currently unknown. These findings should be considered for any child exposed *in utero* to nucleos(t)ide analogues, who present with severe clinical findings of unknown etiology, particularly neurologic findings. These findings do not affect current national recommendations to use antiretroviral therapy in pregnant women to prevent vertical transmission of HIV.

- Immune Reactivation Syndrome**
In HIV infected patients with severe immune deficiency at the time of institution of CART, an inflammatory reaction to asymptomatic or residual opportunistic pathogens may arise and cause serious clinical conditions, or aggravation of symptoms. Typically, such reactions have been observed within the first few weeks or months of initiation of CART. Relevant examples include cytomegalovirus retinitis, generalised and/or focal mycobacterial infections, and *Pneumocystis jirovecii* pneumonia. Any inflammatory symptoms should be evaluated and treatment instituted when necessary.
Autoimmune disorders (such as Graves' disease and autoimmune hepatitis) have also been reported to occur in the setting of immune reactivation; however, the reported time to onset is more variable, and these events can occur many months after initiation of treatment.
- Patients with HIV-1 harbouring mutations**
TAFICITA should be avoided in antiretroviral-experienced patients with HIV-1 harbouring the K65R mutation (see section 5.1).
- Triple nucleoside therapy**
There have been reports of a high rate of virological failure and of emergence of resistance at an early stage when tenofovir disoproxil was combined with lamivudine and abacavir as well as with lamivudine and didanosine as a once daily regimen. Therefore, the same problems may be seen if TAFICITA is administered with a third nucleoside analogue.

- Opportunistic infections**
Patients receiving TAFICITA or any other antiretroviral therapy may continue to develop opportunistic infections and other complications of HIV infection, and, therefore, should remain under close clinical observation by physicians experienced in the treatment of patients with HIV associated diseases.
- Osteonecrosis**
Although the aetiology is considered to be multifactorial (including corticosteroid use, alcohol consumption, severe immunosuppression, higher body mass index), cases of osteonecrosis have been reported particularly in patients with advanced HIV disease and/or long-term exposure to CART. Patients should be advised to seek medical advice if they experience joint aches and pain, joint stiffness or difficulty in movement.
- Nephrotoxicity**
A potential risk of nephrotoxicity resulting from chronic exposure to low levels of tenofovir due to dosing with tenofovir alafenamide cannot be excluded (see section 5.3).
- Patients with end stage renal disease on chronic haemodialysis**
TAFICITA should generally be avoided but may be used in adults with end stage renal disease (estimated CrCl < 15 mL/min) on chronic haemodialysis if the potential benefits outweigh the potential risks (see section 4.2).
- Co-administration of other medicinal products**
The co-administration of TAFICITA is not recommended with certain anticonvulsants (e.g., carbamazepine, oxcarbazepine, phenobarbital and phenytoin), antimycobacterials (e.g., rifampicin, rifabutin, rifapentine), St. John's wort and HIV protease inhibitors

- (Pls) other than atazanavir, lopinavir and darunavir (see section 4.5).
TAFICITA should not be administered concomitantly with medicinal products containing tenofovir alafenamide, tenofovir disoproxil, emtricitabine, lamivudine or adefovir dipivoxil.
- 4.5 **Interaction with other medicinal products and other forms of interaction**
TAFICITA should not be administered concomitantly with medicinal products containing tenofovir alafenamide, tenofovir disoproxil, emtricitabine, lamivudine or adefovir dipivoxil.
- Emtricitabine**
It is shown that the potential for CYP-mediated interactions involving emtricitabine with other medicinal products is low. Co-administration of emtricitabine with medicinal products that are eliminated by active tubular secretion may increase concentrations of emtricitabine, and/or the co-administered medicinal product. Medicinal products that decrease renal function may increase concentrations of emtricitabine.
- Tenofovir alafenamide**
Tenofovir alafenamide is transported by P-glycoprotein (P-gp) and breast cancer resistance protein (BCRP). Medicinal products that strongly affect P-gp and BCRP activity may lead to changes in tenofovir alafenamide absorption. Medicinal products that induce P-gp activity (e.g., rifampicin, rifabutin, carbamazepine, phenobarbital) are expected to decrease the absorption of tenofovir alafenamide, resulting in decreased plasma concentration of tenofovir alafenamide, which may lead to loss of therapeutic effect of TAFICITA and development of resistance. Co-administration of TAFICITA with other medicinal products that inhibit P-gp and BCRP activity (e.g., cobicistat, ritonavir, cyclosporin) is expected to increase the absorption and plasma concentration of tenofovir alafenamide. Co-administration of tenofovir alafenamide and xanthine oxidase inhibitors (e.g., febuxostat) is not expected to increase systemic exposure to tenofovir *in vivo*.
Tenofovir alafenamide is not an inhibitor of CYP1A2, CYP2B6, CYP2C8, CYP2C9, CYP2C19, or CYP2D6 *in vitro*. It is not an inhibitor or inducer of CYP3A *in vivo*. Tenofovir alafenamide is a substrate of OATP1B1 and OATP1B3 *in vitro*. The distribution of tenofovir alafenamide in the body may be affected by the activity of OATP1B1 and OATP1B3.

- Other interactions**
Tenofovir alafenamide is not an inhibitor of human uridine diphosphate glucuronosyltransferase (UGT) 1A1 *in vitro*. It is not known whether tenofovir alafenamide is an inhibitor of other UGT enzymes. Emtricitabine did not inhibit the glucuronidation reaction of a non-specific UGT substrate.
Interactions between the components of TAFICITA and potential co-administered medicinal products are listed in Table 2 (increase is indicated as “↑”, decrease as “↓”, no change as “↔”).
- Table 2: Interactions between the individual components of TAFICITA and other medicinal products**

Medicinal product ¹	by	Effects on medicinal product levels.	Recommendation concerning co-administration with TAFICITA
ANTI-INFECTIVES			
Antifungals			
Ketoconazole Itraconazole		There is no data available with either of the components of TAFICITA. Co-administration of ketoconazole or itraconazole, which are potent P-gp inhibitors, is expected to increase plasma concentrations of tenofovir alafenamide.	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.
Fluconazole Isavuconazole		There is no data available with either of the components of TAFICITA. Co-administration of fluconazole or isavuconazole may increase plasma concentrations of tenofovir alafenamide.	Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
Antimycobacterials			
Rifabutin Rifampicin Rifapentine		There is no data available with either of the components of TAFICITA. Co-administration of rifampicin, rifabutin, and rifapentine, all of which are P-gp inducers, may decrease tenofovir alafenamide plasma concentrations, which may result in loss of therapeutic effect and development of resistance.	Co-administration of TAFICITA and rifabutin rifampicin, or rifapentine is not recommended.
Anti-hepatitis C virus medicinal products			
Ledipasvir (90 mg once daily)/sofosbuvir (400 mg once daily), emtricitabine (200 mg once daily)/ tenofovir alafenamide (10 mg once daily) ²		Ledipasvir: AUC: ↑ C _{max} : ↑ C _{min} : ↑ Sofosbuvir: AUC: ↑ C _{max} : ↑ Sofosbuvir metabolite GS-331007: AUC: ↑ C _{max} : ↔ C _{min} : ↑ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir alafenamide: AUC: ↔ C _{max} : ↔	No dose adjustment of ledipasvir or sofosbuvir is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
Ledipasvir (90 mg once daily)/sofosbuvir (400 mg once daily), emtricitabine (200 mg once daily)/ tenofovir alafenamide (25 mg once daily) ²		Ledipasvir: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Sofosbuvir: AUC: ↔ C _{max} : ↔ Sofosbuvir metabolite GS-331007: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir alafenamide: AUC: ↑ C _{max} : ↔	No dose adjustment of ledipasvir or sofosbuvir is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
Sofosbuvir (400 mg once daily)/velpatasvir (100 mg once daily), emtricitabine (200 mg once daily)/ tenofovir alafenamide (10 mg once daily) ²		Sofosbuvir: AUC: ↑ C _{max} : ↔ Sofosbuvir metabolite GS-331007: AUC: ↑ C _{max} : ↑ Velpatasvir: AUC: ↑ C _{max} : ↑ C _{min} : ↑ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir alafenamide: AUC: ↔ C _{max} : ↓	No dose adjustment of sofosbuvir, velpatasvir or voxilaprevir is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).

Sofosbuvir/velpatasvir/voxilaprevir (400 mg/100 mg/100 mg+100 mg once daily)/ emtricitabine (200 mg once daily)/ tenofovir alafenamide (10 mg once daily) ²	Sofosbuvir: AUC: ↔ C _{max} : ↑ Sofosbuvir metabolite GS-331007: AUC: ↑ C _{max} : ↔ Velpatasvir: AUC: ↔ C _{max} : ↑ C _{min} : ↔ Voxilaprevir: AUC: ↑ C _{max} : ↑ C _{min} : ↑ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir alafenamide: AUC: ↔ C _{max} : ↓ C _{min} : ↓	
Sofosbuvir/velpatasvir/voxilaprevir (400 mg/100 mg/100 mg+100 mg once daily)/ emtricitabine (200 mg once daily)/ tenofovir alafenamide (25 mg once daily) ³	Sofosbuvir: AUC: ↔ C _{max} : ↔ Sofosbuvir metabolite GS-331007: AUC: ↔ C _{max} : ↔ Velpatasvir: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Voxilaprevir: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Emtricitabine: AUC: ↔ C _{max} : ↔ C _{min} : ↔ Tenofovir alafenamide: AUC: ↑ C _{max} : ↑	No dose adjustment of sofosbuvir, velpatasvir or voxilaprevir is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).

ANTIRETROVIRALS			
HIV protease inhibitors			
Atazanavir/cobicistat (300 mg/150 mg once daily), tenofovir alafenamide (10 mg)	Tenofovir alafenamide: AUC: ↑ C _{max} : ↑ Atazanavir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.	
Atazanavir/ritonavir (300/100 mg once daily), tenofovir alafenamide (10 mg)	Tenofovir alafenamide: AUC: ↑ C _{max} : ↑ Atazanavir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.	
Darunavir/cobicistat (800/150 mg once daily), tenofovir alafenamide (25 mg once daily) ⁴	Tenofovir alafenamide: AUC: ↔ C _{max} : ↔ Darunavir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.	
Darunavir/ritonavir (800/100 mg once daily), tenofovir alafenamide (10 mg once daily)	Tenofovir alafenamide: AUC: ↔ C _{max} : ↔ Tenofovir: AUC: ↑ C _{max} : ↑ Darunavir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.	
Lopinavir/ritonavir (800/200 mg once daily), tenofovir alafenamide (10 mg once daily)	Tenofovir alafenamide: AUC: ↑ C _{max} : ↑ Lopinavir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of Emtricitabine and Tenofovir Alafenamide is 200/10 mg once daily.	
Tipranavir/ritonavir	There is no data available with either of the components of TAFICITA. Tipranavir/ritonavir results in P-gp induction. Tenofovir alafenamide exposure is expected to decrease when tipranavir/ritonavir is used in combination with TAFICITA.	Co-administration with TAFICITA is not recommended.	
Other protease inhibitors	Effect is unknown.	There are no data available to make dosing recommendations for co-administration with other protease inhibitors.	
Other HIV antiretrovirals			
Dolutegravir (50 mg once daily), tenofovir alafenamide (10 mg once daily) ²	Tenofovir alafenamide: AUC: ↔ C _{max} : ↔ Dolutegravir: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of TAFICITA is 200/25 mg once daily.	
Rilpivirine (25 mg once daily), tenofovir alafenamide (25 mg once daily)	Tenofovir alafenamide: AUC: ↔ C _{max} : ↔ Rilpivirine: AUC: ↔ C _{max} : ↔ C _{min} : ↔	The recommended dose of TAFICITA is 200/25 mg once daily.	

Note:
Barcodes Should be readable in both the side of the folded leaflet. It should be scannable. Position of barcode can be changed by vendor to meet the above requirement.

				Date of Final	dd / mm / yy	Date of Issue	dd / mm / yy	Page Count	1 of 2
Date/Time of Creation	05/07/22 10:50	Track-Wise PR No.	NA	Client Market	Malaysia	Software / Ver. No.	Adobe Illustrator		
Vendor Job No.	NA	Affiliate New Code	NA	Product Description	LIT TAFICITA (ETA) TABS 200/25MG MY				
Artwork Proof No.	NA	Aft. Superseded Code	NA	New Material Code	75084269	Actual A/w Size	Flat - 340 x 400 mm		
Pharma Code	NA	Barcode Information	NA	ITF Barcode		Other Sizes (if any)	Folded - 34 x 50 mm		
Product Render ID No.	NA	MA Number		Superseded Code	NIL	Pitch After Every	NA		
Main Font Type - 1	Swis721 Cn BT	SAP or SPC Code		Component Type	Printed Literature	Eye / Sensor Mark	NA		
Main Font Type - 2		Packaging Site	Mylan Laboratories Ltd., FDF 3 (P)hampur, Dhar Dist., M.P. - IN	No. of Printed Colors	1	ITF Barcode Size	NA		
Min. Font Size	6pt	Mfg. Lic. No. / Code No.	NA	Printed Colors CMYK / PMS Coated / PMS Color Bridge Coated	BLACK 100%	NA	NA	NA	NA
Reason for Revision / Issuance	New Component			Non Printed Colors	Die Line				
				Equivalent with CMYK (Pantones Ref. Code)	NA				
				Material of Construction	60gsm century maplitpho paper				
Artwork Implementation Schedule, (✓) whichever is applicable	New Component			Design & Style	Supply in folded form as proposed size with tape				
	Immediately (Stock of superseded component to be destroyed, if applicable)								
	After Consumption of Existing (superseded) stock								
	Others (specify)								
LABEL CONTROL / BUSINESS DEVELOPMENT / REGULATORY / MARKETING				Prepared By	Checked By	Approved By			
Sign Offs	Digital Signature			Packaging Technical Services		Production	Regulatory Affairs	Quality Assurance	
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				Date / Remarks		Date / Remarks			

Elavirenz (600 mg once daily), tenofovir alafenamide (40 mg once daily) ¹	Tenofovir alafenamide: AUC: ↓ C _{max} : ↓	The recommended dose of TAFICITA is 200/25 mg once daily.
Maraviroc Nevirapine Raltegravir	There is no data available with either of the components of TAFICITA. Tenofovir alafenamide exposure is not expected to be affected by maraviroc, nevirapine or raltegravir, nor is it expected to affect the metabolic and excretion pathways relevant to maraviroc, nevirapine or raltegravir.	The recommended dose of TAFICITA is 200/25 mg once daily.
ANTICONVULSANTS		
Oxcarbazepine Phenobarbital Phenytoin	There is no data available with either of the components of TAFICITA. Co-administration of oxcarbazepine, phenobarbital, or phenytoin, all of which are P-gp inducers, may decrease tenofovir alafenamide plasma concentrations, which may result in loss of therapeutic effect and development of resistance.	Co-administration of TAFICITA and oxcarbazepine, phenobarbital or phenytoin is not recommended.
Carbamazepine (titrated from 100 mg to 300 mg twice a day), emtricitabine/tenofovir alafenamide (200 mg/25 mg once daily) ^{1,5}	Tenofovir alafenamide: AUC: ↓ C _{max} : ↓ Co-administration of carbamazepine, a P-gp inducer, decreases tenofovir alafenamide plasma concentrations, which may result in loss of therapeutic effect and development of resistance.	Co-administration of TAFICITA and carbamazepine is not recommended.
ANTIDEPRESSANTS		
Sertraline (50 mg once daily), tenofovir alafenamide (10 mg once daily) ²	Tenofovir alafenamide: AUC: ↔ C _{max} : ↔ Sertraline: AUC: ↑ C _{max} : ↑	No dose adjustment of sertraline is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
HERBAL PRODUCTS		
St. John's wort (<i>Hypericum perforatum</i>)	There is no data available with either of the components of TAFICITA. Co-administration of St. John's wort, a P-gp inducer, may decrease tenofovir alafenamide plasma concentrations, which may result in loss of therapeutic effect and development of resistance.	Co-administration of TAFICITA with St. John's wort is not recommended.
IMMUNOSUPPRESSANTS		
Ciclosporin	There is no data available with either of the components of TAFICITA. Co-administration of ciclosporin, a potent P-gp inhibitor, is expected to increase plasma concentrations of tenofovir alafenamide.	The recommended dose of TAFICITA is 200/10 mg once daily.
ORAL CONTRACEPTIVES		
Norgestimate (0.180/0.215/0.250 mg once daily), ethinylestradiol (0.025 mg once daily), emtricitabine/tenofovir alafenamide (200/25 mg once daily) ⁴	Norelgestromin: AUC: ↔ C _{max} : ↔ Ethinylestradiol: AUC: ↔ C _{max} : ↔ Norgestrel: AUC: ↔ C _{max} : ↔ Ethinylestradiol: AUC: ↔ C _{max} : ↔	No dose adjustment of norgestimate/ethinylestradiol is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
SEDATIVES/HYPNOTICS		
Orally administered midazolam (2.5 mg single dose), tenofovir alafenamide (25 mg once daily)	Midazolam: AUC: ↔ C _{max} : ↔	No dose adjustment of midazolam is required. Dose TAFICITA according to the concomitant antiretroviral (see section 4.2).
Intravenously administered midazolam (1 mg single dose), tenofovir alafenamide (25 mg once daily)	Midazolam: AUC: ↔ C _{max} : ↔	

¹When doses are provided, they are the doses used in the data available
² Data using elvitegravir/cobicistat/emtricitabine/tenofovir alafenamide fixed-dose combination tablet
³ Data using emtricitabine/ritonavir/tenofovir alafenamide fixed-dose combination tablet.
⁴ Data using with Emtricitabine and Tenofovir Alafenamide
⁵ Emtricitabine/tenofovir alafenamide was taken with food in this study
⁶ Data using additional voxilaprevir 100 mg to achieve voxilaprevir exposure expected in HCV-infected patients.

4.6 Fertility, pregnancy and lactation
Pregnancy
There are no adequate data on use of TAFICITA or its components in pregnant women. There are no or limited data from the use of tenofovir alafenamide in pregnant women. However, it is shown that use of emtricitabine in pregnant women indicate no malformative nor foetal/neonatal toxicity.
TAFICITA should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus.
Breast-feeding
It is not known whether tenofovir alafenamide is excreted in human milk. Emtricitabine is excreted in human milk. There is insufficient information on the effects of emtricitabine and tenofovir in newborns/infants. Therefore, TAFICITA should not be used during breast-feeding.
In order to avoid transmission of HIV to the infant it is recommended that HIV infected women do not breast-feed their infants under any circumstances.
Fertility
There are no data on fertility from the use of TAFICITA in humans.
4.7 Effects on ability to drive and use machines
Patients should be informed that dizziness has been reported during treatment with TAFICITA.
4.8 Undesirable effects
Summary of the safety profile
The most frequently reported adverse reactions were diarrhoe, nausea and headache.
Tabulated summary of adverse reactions
The adverse reactions in Table 3 are listed by system organ class and frequency. Frequencies are defined as follows: very common, common and uncommon.

Table 3: Tabulated list of adverse reactions	
Frequency	Adverse reaction
Blood and lymphatic system disorders	

Uncommon:	anaemia
Psychiatric disorders	
Common:	abnormal dreams
Nervous system disorders	
Common:	headache, dizziness
Gastrointestinal disorders	
Very common:	nausea
Common:	diarrhoea, vomiting, abdominal pain, flatulence
Uncommon:	dyspepsia
Skin and subcutaneous tissue disorders	
Common:	rash
Uncommon:	Angioedema ^{1,2} , pruritus, urticaria ²
Musculoskeletal and connective tissue disorders	
Uncommon:	arthralgia
General disorders and administration site conditions	
Common:	fatigue

¹ This adverse reaction was identified through post-marketing surveillance for emtricitabine-containing products.
² This adverse reaction was identified through post-marketing surveillance for tenofovir alafenamide-containing products.

Description of selected adverse reactions
Immune Reactivation Syndrome
In HIV infected patients with severe immune deficiency at the time of initiation of CART, an inflammatory reaction to asymptomatic or residual opportunistic infections may arise. Autoimmune disorders (such as Graves' disease and autoimmune hepatitis) have also been reported; however, the reported time to onset is more variable, and these events can occur many months after initiation of treatment (see section 4.4).
Osteonecrosis
Cases of osteonecrosis have been reported, particularly in patients with generally acknowledged risk factors, advanced HIV disease or long-term exposure to CART. The frequency of this is unknown (see section 4.4).
Changes in lipid laboratory tests
In treatment-naïve patients, increases from baseline were observed in both the tenofovir alafenamide fumarate and tenofovir disoproxil fumarate containing treatment groups for the fasting lipid parameters total cholesterol, direct low-density lipoprotein (LDL)- and high-density lipoprotein (HDL)-cholesterol, and triglycerides
Metabolic parameters
Weight and levels of blood lipids and glucose may increase during antiretroviral therapy (see section 4.4).
Paediatric population
The safety profile of emtricitabine and tenofovir alafenamide given with elvitegravir and cobicistat in adolescent patients was similar to that in adults (see section 5.1).
Other special populations
Patients with renal impairment
The safety profile in patients with mild to moderate renal impairment was similar to that in patients with normal renal function (see section 5.1).
There were no new safety issues identified in patients with end stage renal disease on chronic haemodialysis receiving emtricitabine and tenofovir alafenamide, in combination with elvitegravir and cobicistat as a fixed-dose combination tablet (see section 5.2).
Patients co-infected with HIV and HBV
The safety profile of emtricitabine and tenofovir alafenamide in combination with elvitegravir and cobicistat as a fixed-dose combination tablet, in patients with HIV/HBV co-infection, was similar to that in patients with HIV-1 mono-infection (see section 4.4).
4.9 Overdose
If overdose occurs the patient must be monitored for evidence of toxicity (see section 4.8). Treatment of overdose with TAFICITA consists of general supportive measures including monitoring of vital signs as well as observation of the clinical status of the patient. Emtricitabine can be removed by haemodialysis, which removes approximately 30% of the emtricitabine dose over a 3 hour dialysis period starting within 1.5 hours of emtricitabine dosing. Tenofovir is efficiently removed by haemodialysis with an extraction coefficient of approximately 54%. It is not known whether emtricitabine or tenofovir can be removed by peritoneal dialysis.

5. PHARMACOLOGICAL PROPERTIES
5.1 Pharmacodynamic properties
Pharmacotherapeutic group: Antiviral for systemic use; antivirals for treatment of HIV infections, combinations. ATC code: J05AR17.
Mechanism of action
Emtricitabine is a nucleoside reverse transcriptase inhibitor (NRTI) and nucleoside analogue of 2'-deoxycytidine. Emtricitabine is phosphorylated by cellular enzymes to form emtricitabine triphosphate. Emtricitabine triphosphate inhibits HIV replication through incorporation into viral deoxyribonucleic acid (DNA) by the HIV reverse transcriptase (RT), which results in DNA chain-termination. Emtricitabine has activity against HIV-1, HIV-2, and HBV.
Tenofovir alafenamide is a nucleotide reverse transcriptase inhibitor (NRTI) and phosphonamidate prodrug of tenofovir (2'-deoxyadenosine monophosphate analogue). Tenofovir alafenamide is permeable into cells and due to increased plasma stability and intracellular activation through hydrolysis by cathepsin A, tenofovir alafenamide is more efficient than tenofovir disoproxil fumarate in concentrating tenofovir in peripheral blood mononuclear cells (PBMCs) or HIV target cells including lymphocytes and macrophages. Intracellular tenofovir is subsequently phosphorylated to the pharmacologically active metabolite tenofovir diphosphate. Tenofovir diphosphate inhibits HIV replication through incorporation into viral DNA by the HIV RT, which results in DNA chain-termination. Tenofovir has activity against HIV-1, HIV-2, and HBV.
Antiviral activity
Emtricitabine and tenofovir alafenamide demonstrated synergistic antiviral activity in cell culture. No antagonism was observed with emtricitabine or tenofovir alafenamide when combined with other antiretroviral agents.

Resistance
In treatment-naïve patients
Most HIV-1 isolates from patients who developed resistance mutations to elvitegravir in integrase also developed resistance mutations to emtricitabine in RT.
In patients co-infected with HIV and HBV
No amino acid substitutions associated with resistance to any of the components of E/C/F/TAF were identified in HIV-1 or HBV.
Cross-resistance in HIV-1 infected, treatment-naïve or virologically suppressed patients
Emtricitabine-resistant viruses with the M184V/I substitution were cross-resistant to lamivudine, but retained sensitivity to didanosine, stavudine, tenofovir, and zidovudine. The K65R and K70E mutations result in reduced susceptibility to abacavir, didanosine, lamivudine, emtricitabine, and tenofovir, but retain sensitivity to zidovudine. Multinucleoside-resistant HIV-1 with a T69S double insertion mutation or with a Q151M mutation complex including K65R showed reduced susceptibility to tenofovir alafenamide

5.2 Pharmacokinetic properties
Absorption
Emtricitabine is rapidly and extensively absorbed following oral administration with peak plasma concentrations occurring at 1 to 2 hours post-dose. Following multiple dose oral administration of emtricitabine, the (mean ± SD) steady state plasma emtricitabine peak concentrations (C_{max}) were 1.8 ± 0.7 µg/mL and the area-under the plasma concentration-time curve over a 24-hour dosing interval (AUC) was 10.0 ± 3.1 µg.h/mL. The mean steady state plasma trough concentration at 24 hours post-dose was equal to or greater than the mean IC90 value for anti-HIV-1 activity.
Emtricitabine systemic exposure was unaffected when emtricitabine was administered with food.
Following administration of food, peak plasma concentrations were observed approximately 1 hour post-dose for tenofovir alafenamide administered as emtricitabine/tenofovir alafenamide (25 mg) or elvitegravir/cobicistat/emtricitabine/tenofovir alafenamide fumarate (10 mg). The mean C_{max} and AUC₀₋₂₄ (mean ± SD) under fed conditions following a single 25 mg dose of tenofovir alafenamide administered in TAFICITA were 0.21 ± 0.13 µg/mL and 0.25 ± 0.11 µg.h/mL, respectively. The mean C_{max} and AUC₀₋₂₄ following a single 10 mg dose of tenofovir alafenamide administered in elvitegravir/cobicistat/emtricitabine/tenofovir alafenamide fumarate were 0.21 ± 0.10 µg/mL and 0.25 ± 0.08 µg.h/mL, respectively.
Relative to fasting conditions, the administration of tenofovir alafenamide with a high fat meal (~800 kcal, 50% fat) resulted in a decrease in tenofovir alafenamide C_{max} (15-37%) and an increase in AUC₀₋₂₄ (17-77%).
Distribution
Binding of emtricitabine to human plasma proteins was < 4% and independent of concentration over the range of 0.02-200 µg/mL. At peak plasma concentration, the mean plasma to blood drug concentration ratio was ~1.0 and the mean semen to plasma drug concentration ratio was ~4.0.
Binding of tenofovir to human plasma proteins is < 0.7% and is independent of concentration over the range of 0.01-25 µg/mL. Binding of tenofovir alafenamide to human plasma proteins was approximately 80%.
Biotransformation
It is indicated that emtricitabine is not an inhibitor of human CYP enzymes. Following administration of [¹⁴C]-emtricitabine, complete recovery of the emtricitabine dose was achieved in urine (~86%) and faeces (~14%). Thirteen percent of the dose was recovered in the urine as three putative metabolites. The biotransformation of emtricitabine includes oxidation of the thiol moiety to form the 3'-sulfoxide diastereomers (~9% of dose) and conjugation with glucuronic acid to form 2'-O-glucuronide (~4% of dose). No other metabolites were identifiable.
Metabolism is a major elimination pathway for tenofovir alafenamide in humans, accounting for > 80% of an oral dose. It is shown that tenofovir alafenamide is metabolised to tenofovir (major metabolite) by cathepsin A in PBMCs (including lymphocytes and other HIV target cells) and macrophages; and by carboxylesterase-1 in hepatocytes. Tenofovir alafenamide is hydrolysed within cells to form tenofovir (major metabolite), which is phosphorylated to the active metabolite tenofovir diphosphate. A 10 mg oral dose of tenofovir alafenamide (given with emtricitabine and elvitegravir and cobicistat) resulted in tenofovir diphosphate concentrations > 4-fold higher in PBMCs and > 90% lower concentrations of tenofovir in plasma as compared to a 245 mg oral dose of tenofovir disoproxil (as fumarate) (given with emtricitabine and elvitegravir and cobicistat).
Tenofovir alafenamide is not metabolised by CYP1A2, CYP2C8, CYP2C9, CYP2C19, or CYP2D6. Tenofovir alafenamide is minimally metabolised by CYP3A4. Upon co-administration with the moderate CYP3A inducer efavirenz, tenofovir alafenamide exposure

was not significantly affected. Following administration of tenofovir alafenamide, plasma [¹⁴C]-radioactivity showed a time-dependent profile with tenofovir alafenamide as the most abundant species in the initial few hours and uric acid in the remaining period.
Elimination
Emtricitabine is primarily excreted by the kidneys with complete recovery of the dose achieved in urine (approximately 86%) and faeces (approximately 14%). Thirteen percent of the emtricitabine dose was recovered in urine as three metabolites. The systemic clearance of emtricitabine averaged 307 mL/min. Following oral administration, the elimination half-life of emtricitabine is approximately 10 hours.
Renal excretion of intact tenofovir alafenamide is a minor pathway with < 1% of the dose eliminated in urine. Tenofovir alafenamide is mainly eliminated following metabolism to tenofovir. Tenofovir alafenamide and tenofovir have a median plasma half-life of 0.51 and 32.37 hours, respectively. Tenofovir is renally eliminated by both glomerular filtration and active tubular secretion.
Pharmacokinetics in special populations
Age, gender, and ethnicity
No clinically relevant pharmacokinetic differences due to age, gender or ethnicity have been identified for emtricitabine, or tenofovir alafenamide.
Paediatric population
Exposures of emtricitabine and tenofovir alafenamide (given with elvitegravir and cobicistat) achieved in paediatric patients aged 12 to < 18 years who received emtricitabine and tenofovir alafenamide given with elvitegravir and cobicistat were similar to exposures achieved in treatment-naïve adults (Table 4).

Table 4: Pharmacokinetics of emtricitabine and tenofovir alafenamide in antiretroviral-naïve adolescents and adults						
	Adolescents			Adults		
	FTC ^a	TAF ^a	TFV ^a	FTC ^a	TAF ^a	TFV ^a
AUC ₀₋₂₄ (ng.h/mL)	14,424.4	242.8	275.8	11,714.1	206.4	292.6
C _{max} (ng/mL)	2,265.0	121.7	14.6	2,056.3	162.2	15.2
C _{min} (ng/mL)	102.4 ^b	N/A	10.0	95.2	N/A	10.6

E/C/F/TAF = elvitegravir/cobicistat/emtricitabine/tenofovir alafenamide fumarate
FTC = emtricitabine; TAF = tenofovir alafenamide fumarate; TFV = tenofovir
N/A = not applicable

Renal impairment
No clinically relevant differences in tenofovir alafenamide, or tenofovir pharmacokinetics were observed between a healthy person and patient with severe renal impairment (estimated CrCl ≥ 15 mL/min and < 30 mL/min). Mean systemic emtricitabine exposure was higher in patients with severe renal impairment (estimated CrCl < 30 mL/min) (33.7 µg.h/mL) than in person with normal renal function (11.8 µg.h/mL). The safety of emtricitabine and tenofovir alafenamide has not been established in patients with severe renal impairment (estimated CrCl ≥ 15 mL/min and < 30 mL/min).
Exposures of emtricitabine and tenofovir in patients with end stage renal disease (estimated CrCl < 15 mL/min) on chronic haemodialysis who received emtricitabine and tenofovir alafenamide in combination with elvitegravir and cobicistat as a fixed-dose combination tablet (E/C/F/TAF) were significantly higher than in patients with normal renal function. No clinically relevant differences in tenofovir alafenamide pharmacokinetics were observed in patients with end stage renal disease on chronic haemodialysis as compared to those with normal renal function. There were no new safety issues in patients with end stage renal disease on chronic haemodialysis receiving emtricitabine and tenofovir alafenamide, in combination with elvitegravir and cobicistat as a fixed-dose combination tablet (see section 4.8).
There are no pharmacokinetic data on emtricitabine or tenofovir alafenamide in patients with end stage renal disease (estimated CrCl < 15 mL/min) not on chronic haemodialysis. The safety of emtricitabine and tenofovir alafenamide has not been established in these patients.
Hepatic impairment
There is no data of pharmacokinetics of emtricitabine in patients with hepatic impairment; however, emtricitabine is not significantly metabolised by liver enzymes, so the impact of liver impairment should be limited.
Clinically relevant changes in the pharmacokinetics of tenofovir alafenamide or its metabolite tenofovir were not observed in patients with mild or moderate hepatic impairment. In patients with severe hepatic impairment, total plasma concentrations of tenofovir alafenamide and tenofovir are lower than those seen in person with normal hepatic function. When corrected for protein binding, unbound (free) plasma concentrations of tenofovir alafenamide in severe hepatic impairment and normal hepatic function are similar.
Hepatitis B and/or hepatitis C virus co-infection
The pharmacokinetics of emtricitabine and tenofovir alafenamide have not been fully evaluated in patients co-infected with HBV and/or HCV.
PHARMACEUTICAL PARTICULARS
6.1 List of excipients
Core Tablet
Microcrystalline cellulose
Croscarmellose sodium
Magnesium stearate

Film coat
Polyvinyl alcohol
Titanium dioxide
Macrogol
Talc
FD&C Blue #2/Indigo carmine aluminium lake

6.2 Incompatibilities
Not applicable.

6.3 Shelf life
24 months.

6.4 Special precautions for storage
Do not store above 30°C, store in the original container.

6.5 Nature and contents of container
HDPE bottle of 30's pack

6.6 Special precautions for disposal
Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

PRODUCT REGISTRATION HOLDER

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		After Consumption of Existing (superseded) stock			Design & Style		Supply in folded form as proposed size with tape		
		Others (specify)			Prepared By		Checked By	Approved By	
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